

Explainable Artificial Intelligence (XAI)

Core Ideas, Techniques, and Solutions

Dwivedi et al., 2023 · ACM Computing Surveys · doi:10.1145/3561048

Citations	Downloads	Published
986	42,370	January 2023

What we cover today

Part 1 (6 min) — The Problem: why ML models need explanation, the black box problem, real consequences of unexplainable AI.

Part 2 (8 min) — The Methods: taxonomy of XAI techniques, SHAP, LIME, and global vs. local explanations.

Part 3 (6 min) — The Takeaway: what XAI actually adds, when it is required, and how to choose the right tool.

Central question throughout: If I already know SHAP and LIME, what does 'XAI' tell me that is genuinely new?

Why accuracy alone is not enough

A model can be 96% accurate and completely wrong about *why*.

The Clever Hans problem	A pneumonia detection model learned to recognize hospital ID tags in the corner of X-rays — not lung tissue. 96% accurate. Completely useless clinically.
Google Images, 2015	Searching 'CEO' returned women only 11% of the time despite real-world representation of 27%. The model learned bias from training data — invisibly.
Loan denial (GDPR)	EU law requires any automated decision affecting a person to be explainable on request. A neural net prediction alone cannot satisfy this — legally.

"Just having a highly accurate model is not sufficient to trust and deploy the model in real-world applications."

— Dwivedi et al., 2023

Three things accuracy metrics miss:

1. **What the model actually learned** — could be spurious correlations, not meaningful patterns.
2. **Whether reasoning is fair** — bias may be invisible in aggregate accuracy metrics.
3. **Why a specific decision was made** — needed for audits, appeals, and regulatory compliance.

Student question: Is XAI just a methodology?

"If I can run a neural network and then run SHAP without knowing XAI — what has XAI told me new?"

1 — Unified vocabulary	Before XAI, SHAP, LIME, and saliency maps were scattered across research communities with no common language. XAI packages them with consistent terms: global vs. local, model-agnostic vs. model-specific, post-hoc vs. intrinsic.
2 — Legal obligation	XAI defines when explanation is legally required — not just technically possible. GDPR, healthcare regulation, financial audits. Running SHAP for curiosity vs. running it for compliance are different acts.
3 — Catches what accuracy misses	XAI is the only framework that systematically checks whether a model's reasoning is valid — not just its output. This is the Clever Hans problem. You cannot find it without an explanation tool.

Short answer: XAI is not a new algorithm. If you already use SHAP on your neural network, you are already doing XAI. What XAI adds is the structured argument that you are *obligated* to do it in certain domains — and the taxonomy of which tool answers which question.

When does explainability actually matter?

Stakes / Context	Requirement	Example domains	What happens without XAI
High stakes · High human involvement	Legally required	Loan approval, medical diagnosis, criminal risk scoring, hiring decisions	GDPR Article 22 violation. Affected persons cannot appeal. Regulator fines.
Medium stakes · Auditability needed	Strongly recommended	Fraud detection, insurance pricing, predictive maintenance	Cannot prove model is unbiased against protected groups during audit.
Any stakes · Development phase	For debugging	Any model before deployment	Clever Hans behavior ships to production undetected.
Low stakes · Low human involvement	Optional	Spam filter, movie recommendation, ad placement	User gets a bad recommendation — recoverable, low consequence.

Taxonomy of XAI techniques

Three axes classify every XAI method: scope · model dependency · transparency

	GLOBAL — whole model behavior	LOCAL — one prediction at a time
WHITE BOX self-explanatory	Linear / Logistic regression · Decision tree · Random forest (Gini) · GAMs · Feature importance · PDP / ICE / ALE	Logistic regression (beta*x) · Decision tree path · Neural-symbolic · Integrated gradients
BLACK BOX needs add-on tool	SHAP global summary · Tree SHAP · Kernel SHAP · Permutation importance · Global surrogate	SHAP waterfall · LIME · Counterfactuals · Anchors · CEM

Key rule: Deep learning (neural nets) always lives in the black box rows. No white-box option exists once hidden layers are added — always requires a post-hoc add-on tool like SHAP or LIME.

SHAP — the workhorse of black-box explanation

Game-theoretic: asks 'what is each feature's fair contribution to this prediction?'

The core idea

For each prediction, SHAP computes the average marginal contribution of a feature across all possible subsets of other features. The result $\phi(i)$ is that feature's 'Shapley value' — its fair share of credit for the prediction.

The additive property

Sum of all $\phi(i) = f(x)$ minus $E[f(x)]$. All SHAP values for one prediction always sum exactly to the prediction's deviation from the dataset mean. This is mathematically guaranteed — unlike LIME.

Variant	Works on	Speed	Accuracy
Tree SHAP	XGBoost, Random Forest, any tree model	Fast (polynomial time)	Exact values
Kernel SHAP	Any model including neural networks	Slower (sampling)	Approximate

Local SHAP — loan denial example

Baseline (average approval probability) = 0.50

debt ratio: +0.22 (pushes toward DENIED — high debt is the biggest driver)

credit score: +0.14 (pushes toward DENIED — low score matters)

loan amount: +0.08 (pushes toward DENIED — large loan is risky)

income: -0.07 (pushes toward APPROVED — partially offsets)

age: -0.02 (pushes toward APPROVED — small positive signal)

Final denial probability = 0.87

LIME — local approximation of any black box

Explains one prediction by building a simple interpretable model in its local neighborhood.

How LIME works — four steps

1. **Take one data point to explain** — the specific instance whose prediction you want to understand (e.g. one loan applicant).
2. **Perturb the input hundreds of times** — slightly modify feature values, run each through the black-box model, and collect predictions.
3. **Fit a local linear model on those samples** — weight samples by proximity to the original point, fit a simple linear regression to approximate local behavior.
4. **Read off the linear coefficients as the explanation** — coefficients of the local linear model = feature contributions for that one prediction.

LIME vs. SHAP — when to use which

	LIME	SHAP
Speed	Faster	Slower for neural nets (Kernel SHAP)
Stability	Can vary between runs on same instance	Consistent
Additivity	Not guaranteed	Always sums to $f(x)$ minus $E[f(x)]$
Best for	Quick interactive exploration	Final reporting, stakeholders, audits

Key limitation of LIME: LIME defines 'neighborhood' arbitrarily. Two runs on the same instance can give different feature weights. Use SHAP for anything you will show a stakeholder or include in a compliance report.

SHAP global feature importance

Aggregate local explanations across the whole dataset to understand overall model behavior.

How it works

Step 1: Compute local SHAP values for every row — this gives you an $N \times F$ matrix of $\phi(i)$ values (instances $\times$ features).

Step 2: Take the mean absolute $\phi(i)$ per feature across all rows.

Step 3: Rank features by that mean absolute value. Larger = more influential to the model overall.

Why absolute value matters:

Income pushes some predictions up (+0.73) and others down (-0.18) depending on the applicant. Its average $|\phi| = 0.45$ still correctly ranks it as an important feature. Signs cancel; magnitude is preserved.

Example global ranking (loan model):

Rank	Feature	Mean SHAP	Interpretation
1	income	0.45	Strongest overall driver of approval/denial
2	debt ratio	0.44	Near-equal importance to income
3	credit score	0.31	Significant but secondary
4	loan amount	0.23	Moderate effect
5	age	0.14	Minor influence — good sign for fairness audit

SHAP vs. `sklearn.feature_importances_`: Built-in Gini importance is biased toward high-cardinality features. Tree SHAP gives unbiased, exact values. Always prefer SHAP for final reporting.

Student question: which method do I run and when?

"After running a neural network, to answer the question of reasoning — SHAP, LIME, or something else?"

Your model	Your question	Use this	Why
Neural net	Which features matter overall?	Kernel SHAP — global	No built-in importance; SHAP is model-agnostic
Neural net	Why was THIS prediction made?	SHAP waterfall	Additive, stable, auditable — use for stakeholders and compliance
Neural net	Quick exploration / what-if?	LIME	Faster; but unstable so never use for final reporting
XGBoost / RF	Which features matter overall?	Tree SHAP — global	Exact, fast — do NOT use sklearn .feature_importances_
XGBoost / RF	Why was THIS prediction made?	Tree SHAP — local waterfall	Exact Shapley values in polynomial time on trees
Logistic reg.	Which features matter overall?	Coefficients directly	White box — beta IS the explanation. No add-on needed.
Logistic reg.	Why was THIS prediction made?	beta * x per instance	Multiply coefficient x feature value per row. Exact, no tool needed.
Any model	Is the model biased / fair?	SHAP beeswarm	Check if protected attributes rank high in mean $ \phi(i) $

Rule of thumb: SHAP for anything that goes to a stakeholder or audit. LIME for quick interactive exploration only. Coefficients if you are still on a white-box model — no add-on needed.

What this paper actually contributes

Not a new algorithm	XAI does not invent SHAP, LIME, or PDP. These already existed. XAI is the framework that organizes them, names the problem they solve, and defines when you are obligated to use them.
A taxonomy you can use	Global vs. local, white box vs. black box, model-specific vs. agnostic. These three axes let you navigate from 'I have this model and this question' to 'I should use this tool' without guessing.
Accuracy does not equal trustworthiness	High accuracy is necessary but not sufficient for deployment. XAI provides the second check: is the reasoning valid? The Clever Hans example shows a 96% accurate model can be entirely wrong about what it learned.

"Model performance metrics do not always exhibit true prediction decisions. Just having a highly accurate model is not sufficient to trust and deploy the model."

— Dwivedi et al., 2023

The three things XAI gives a data scientist:

1. **Debugging:** find Clever Hans behavior before deployment by examining what the model actually learned.
2. **Compliance:** satisfy GDPR and domain-specific audit requirements by producing auditable explanation outputs.
3. **Communication:** explain model decisions to non-technical stakeholders, regulators, and affected users.

Open questions for the group

- Q1** XAI is described as a 'methodology, not a new algorithm.' Do you think that framing understates or overstates its contribution to the field?
- Q2** SHAP values tell you which features drove a prediction — but they don't tell you if those features are ethically appropriate to use. Where does XAI end and ethical AI begin?
- Q3** There is a known accuracy-interpretability tradeoff. Should regulators mandate explainability even if it means deploying a less accurate model in healthcare or finance?
- Q4** With LLMs now making decisions in real applications, do you think SHAP and LIME are sufficient XAI tools — or does the field need entirely new explanation methods for generative AI?